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CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
Acceleration due to gravity	g	=	9.8 m s^{-2}
1 Atmosphere	Atm	=	1.00×10^5 N m^{-2}
Boltzmann's constant	k	=	1.38×10^{-23} J K^{-1}
Density of water		=	1.00×10^3 kg m^{-3}
Specific heat capacity of water		=	4200 J kg^{-1} K^{-1}
Specific latent heat of fusion of ice		=	3.34×10^5 J kg^{-1}
Specific latent heat of vaporization of water		=	2.26×10^6 J kg^{-1}
Avogadro's constant	N_A	=	6.02×10^{23} per mole
Molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Stefan-Boltzmann's constant	s	=	5.67×10^{-8} W m^{-2} K^{-4}
Speed of light in free space	c	=	3.00×10^8 m s^{-1}

GO ON TO THE NEXT PAGE

SECTION A

Attempt ALL questions.

You MUST write your answers in this answer booklet.

1. (a) Table 1 shows the data collected in a terminal velocity experiment. A small lead sphere of mass m and radius r was timed as it fell through glycerine contained in a long tube at 30 °C.

Table 1

Time, t/s	Velocity, V/m s^{-1}
0	0
0.4	2.53
0.8	3.51
1.2	3.89
1.6	4.03
2.0	4.09
2.4	4.11
2.8	4.12
3.0	4.12

(i) On the grid on page 5, plot a graph of velocity, V, against time, t. **[5 marks]**

(ii) Explain the shape of the graph and use it to identify the terminal velocity, V_t, of the sphere.

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[2 marks]

(iii) Determine the average acceleration of the sphere between $t = 0.5$ s and $t = 0.7$ s.

[2 marks]

(b) The terminal velocity, V_t, of the lead sphere, is given by

$$V_t = \frac{mg}{6\pi kr}$$

where k is a temperature-dependent constant that determines the resistance to motion in the fluid.

(i) Determine the units of k.

[3 marks]

(ii) Given that $m = 5 \times 10^{-3}$ kg, and $r = 1 \times 10^{-3}$ m, determine the value of k for glycerine at 30 °C.

[2 marks]

(iii) Explain how the terminal velocity will be affected if a sphere of the same mass but twice the radius is used.

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[1 mark]

Total 15 marks

2. (a) Explain how a stringed instrument such as a guitar produces a musical note.

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[2 marks]

(b) The apparatus shown in Figure 1 may be used to investigate waves on strings.

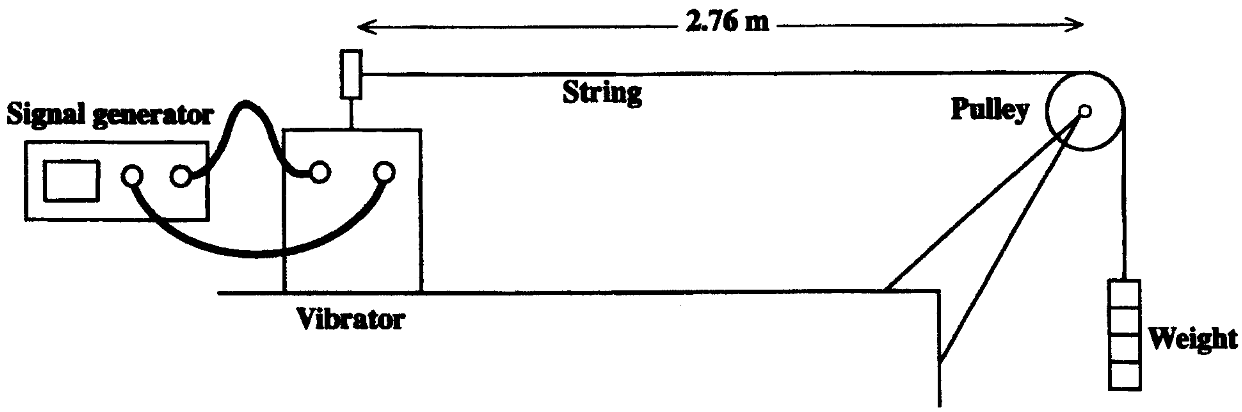

Figure 1

(i) By varying the setting on the signal generator a standing wave with 3 anti-nodes may be set up on the string. In the space below draw a diagram to show how the string would look when this standing wave is set up. (Note: There is no need to draw the vibrator or pulley.)

[1 mark]

(ii) Calculate the wavelength of the wave you have drawn.

[1 mark]

(iii) Write an equation for the wavelength when the string has n anti-nodes and use this to show that the relationship between the frequency of the vibrator and the number of anti-nodes is $f = \frac{v}{2L} n$

where v is the wave velocity and L (= 2.76 m) is the length of the string.

[3 marks]

(c) The data in the Table 2 were collected using the apparatus in Figure 1 on page 7. By means of plotting a suitable graph on page 8, find the velocity of the waves on the string.

[5 marks]

Table 2

***f*/Hz**	8.6	16	25	35	42	52	62	69
No. of anti-nodes, n	1	2	3	4	5	6	7	8

<u>Calculations</u>

[3 marks]

Total 15 marks

3. (a) List ONE advantage and ONE disadvantage of using a liquid in glass thermometer, a thermocouple and a constant volume gas thermometer to measure temperature.

Themometer	Advantage	Disadvantage
Liquid in glass thermometer		
Thermocouple		
Constant volume gas thermometer		

[6 marks]

(b) Figure 2 shows the setup of the experimental arrangement to determine the boiling point of a liquid. The following apparatus is available: a constant volume gas thermometer, large beaker, electric heater, ice, distilled water, liquid, ruler and a stirrer.

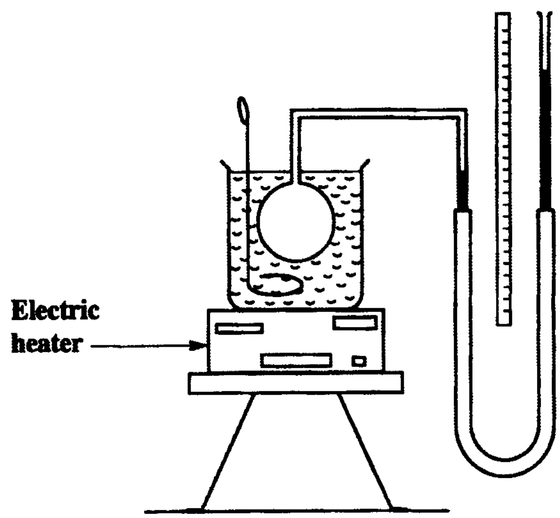

Figure 2

The boiling point, t, of the liquid on the empirical centigrade scale is given by

$$t = 100 \frac{h_t - h_o}{h_{100} - h_o}$$

where h_o is the height of the mercury column at 0 °C, h_{100} is the height of the mercury column at 100 °C and h_t, the height of the mercury column at t °C.

(i) How would you ensure that the volume of gas in the bulb is held constant?

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[1 mark]

(ii) Carefully explain how readings are taken to determine

a) h_0, the height of the mercury column at 0 °C

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b) h_{100}, the height of the mercury column at 100 °C

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c) h_t, the height of the mercury column at t °C.

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d) Indicate on the diagram (Figure 2) where the height, h, is to be measured.

[4 marks]

(c) The following results were obtained:

$h_o = 5.0$ cm

$h_{100} = 20.0$ cm

$h_t = 16.8$ cm

(i) Using the above results, determine the boiling point of the liquid.

[1 mark]

(ii) Determine the pressure of the gas in the bulb when the liquid is at its boiling point.

Density of mercury 13 600 kg m^{-3}

Atmospheric pressure, 76 cm Hg

Acceleration due to gravity, $g = 9.81$ m s^{-2}

[3 marks]

Total 15 marks

SECTION B

Attempt ALL questions.

You MUST write your answers in the answer booklet provided.

4. (a) A body of mass, m, is moving in a circle of radius, r, with constant speed v.

   (i) Explain why there must be an acceleration experienced by the mass although it is moving at constant speed.

   (ii) Write an expression for the magnitude of the acceleration and state the direction of this acceleration.

   (iii) Explain why the work done by the centripetal force on the mass is zero.

[5 marks]

(b) Figure 3 shows a pendulum with string of length 0.5 m and mass 1 kg being whirled at constant speed in a horizontal circle. The mass is 1.5 m above the ground.

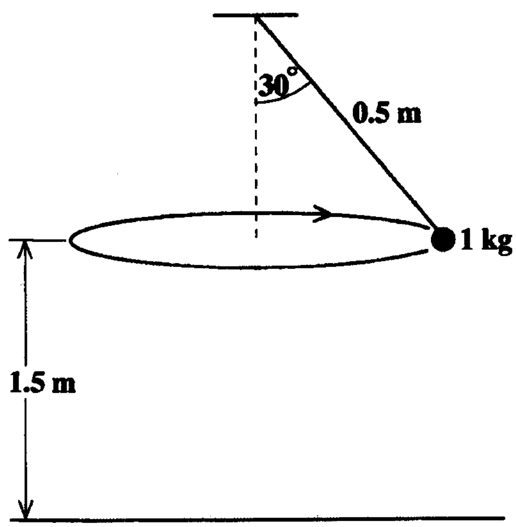


Figure 3

   (i) Draw the free body diagram showing the forces acting on the mass.

   (ii) Calculate

      a) the tension in the string

      b) the speed of the mass. **[6 marks]**

(c) During the motion the string suddenly breaks.

   (i) Describe the subsequent motion of the mass.

   (ii) Calculate the time it takes for the mass to hit the ground after the string breaks.

[4 marks]

Total 15 marks

5. (a) A diffraction grating is illuminated by a parallel beam of light with a mixture of three wavelengths in the yellow, blue and red parts of the visible spectrum as shown in Figure 4.

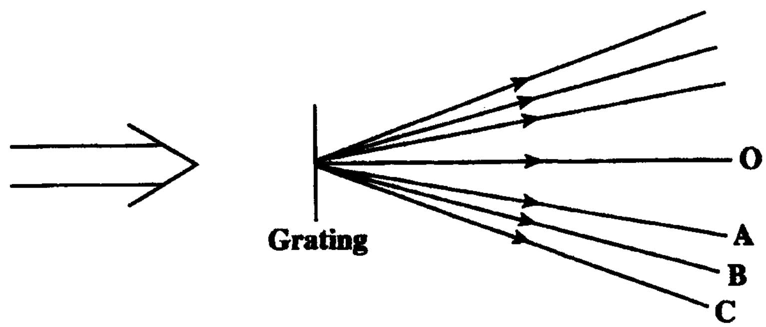


Figure 4

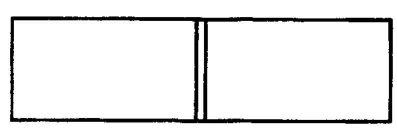
(i) Discuss with the aid of suitable diagrams the role of *diffraction* and *interference* of waves in the production of this spectrum.

(ii) Explain why Beam A in Figure 4 must be the blue light and also identify the colours of the Beams B and C.

(iii) Why does the central beam (labelled O) contain a mixture of all three colours?

[8 marks]

(b) The spectrum of sodium contains two yellow lines close together with wavelengths of **589** nm and **590** nm as shown in Figure 5.



Sodium D lines

Figure 5

To view these lines separately an experimenter used a diffraction grating with **6×10^5** lines per metre and the highest order spectrum possible.

(i) Calculate the value of the HIGHEST order spectrum which can be used to view the sodium lines.

(ii) What is the angular separation of the two sodium lines in this spectrum?

[7 marks]

Total 15 marks

6. (a) The first law of thermodynamics is given by the equation

$$\Delta U = Q + W.$$

(i) Explain the meaning of EACH of the terms used in the equation when the law is applied to the heating of a fixed mass of gas.

(ii) Use the first law of thermodynamics to explain why the molar heat capacity at constant pressure, C_p, is greater than the molar heat capacity at constant volume, C_v. **[6 marks]**

(b) One mole of an ideal monatomic gas, $C_v = \frac{3}{2}R$, is taken through the cycle represented by states 1 to 4 as shown in Figure 6.

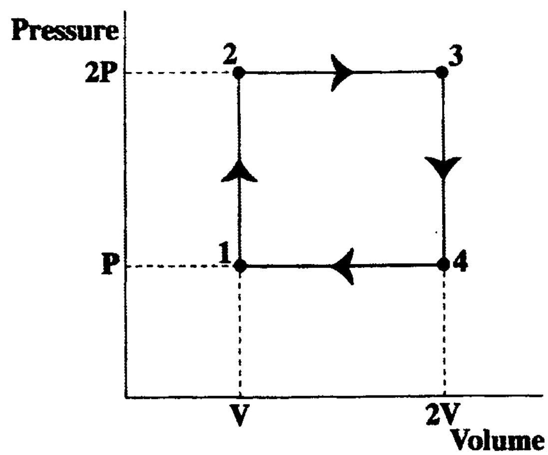


Figure 6

Assume that at state 1, $P_1 = 1.01 \times 10^5$ Pa, $V_1 = 0.0225$ m^3 and $T_1 = 273$ K and at state 3, $T_3 = 1092$ K.

Calculate

(i) the work done during the cycle

(ii) the temperature T_2 at state 2

(iii) the energy added as heat during the processes $1 \rightarrow 2$ and $2 \rightarrow 3$

(iv) the efficiency of the cycle.

[9 marks]

Total 15 marks

END OF TEST